

SenseQuiet: Intelligent Noise Monitoring & Alert System

Custom Analog Signal Conditioning, LabVIEW Integration, and Decoupled UI



PROBLEM STATEMENT

The Case for Automated Noise Enforcement

Subjective & Inconsistent

Human monitoring in libraries and hospitals is inherently unreliable – enforcement varies by individual and time of day.

Expensive Black-Box Solutions

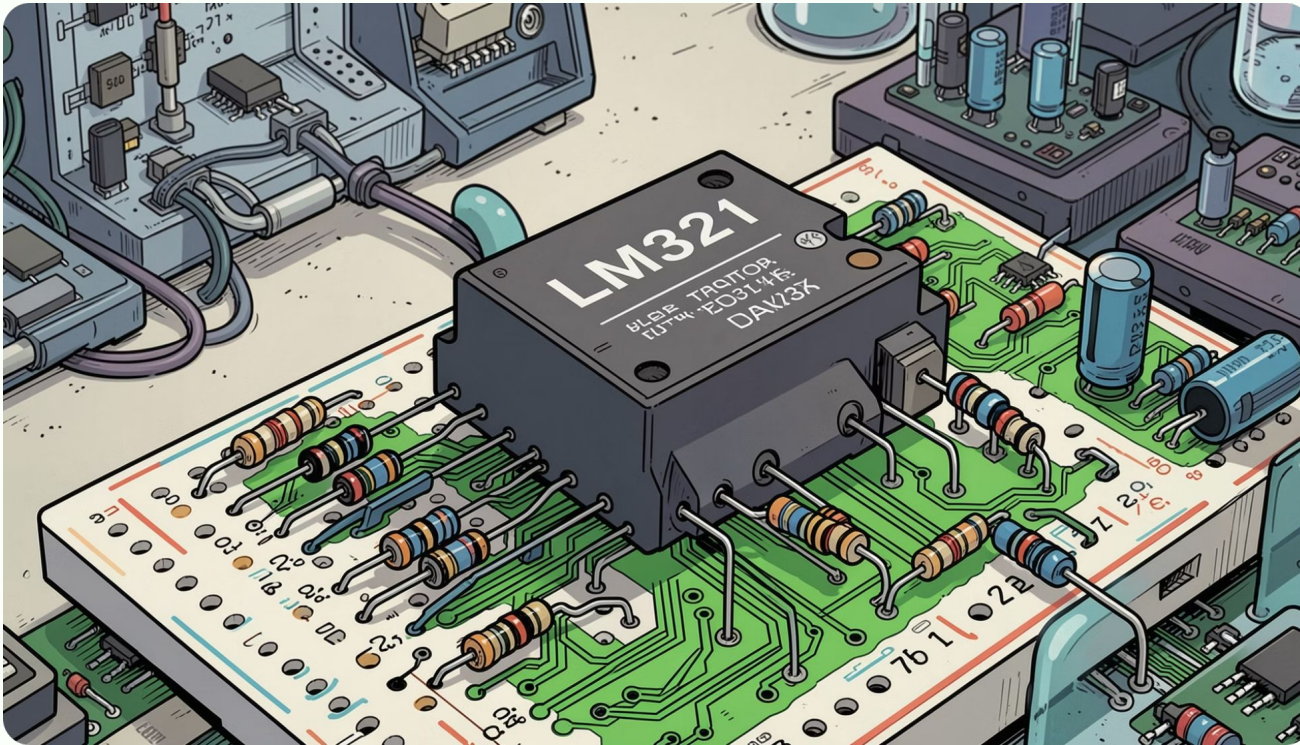
Commercial acoustic sensors are costly, proprietary, and offer no transparency into their signal processing pipelines.

Health Impact Requires Action

Prolonged exposure to elevated decibel levels is a documented occupational hazard. Objective, automated enforcement is essential.



Custom Analog Sound Sensor — Amplification Stage



Electret Microphone Front-End

An electret capsule captures acoustic energy, producing a millivolt-level AC signal too weak for direct DAQ acquisition.

LM324 Non-Inverting Amplifier

A single-supply LM324 Op-Amp (5 V rail) provides gain. A **2.5 V virtual ground** – established via a resistor divider and bypass capacitor – biases the signal to prevent negative-cycle clipping.

Custom Analog Sound Sensor — Conditioning Stage

Envelope Detection

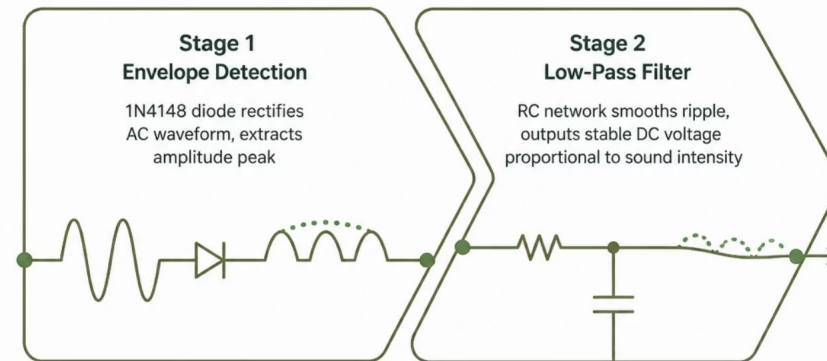
The amplified AC waveform passes through a **1N4148 fast-switching diode** envelope detector, extracting the signal's amplitude envelope.

Low-Pass Filtering

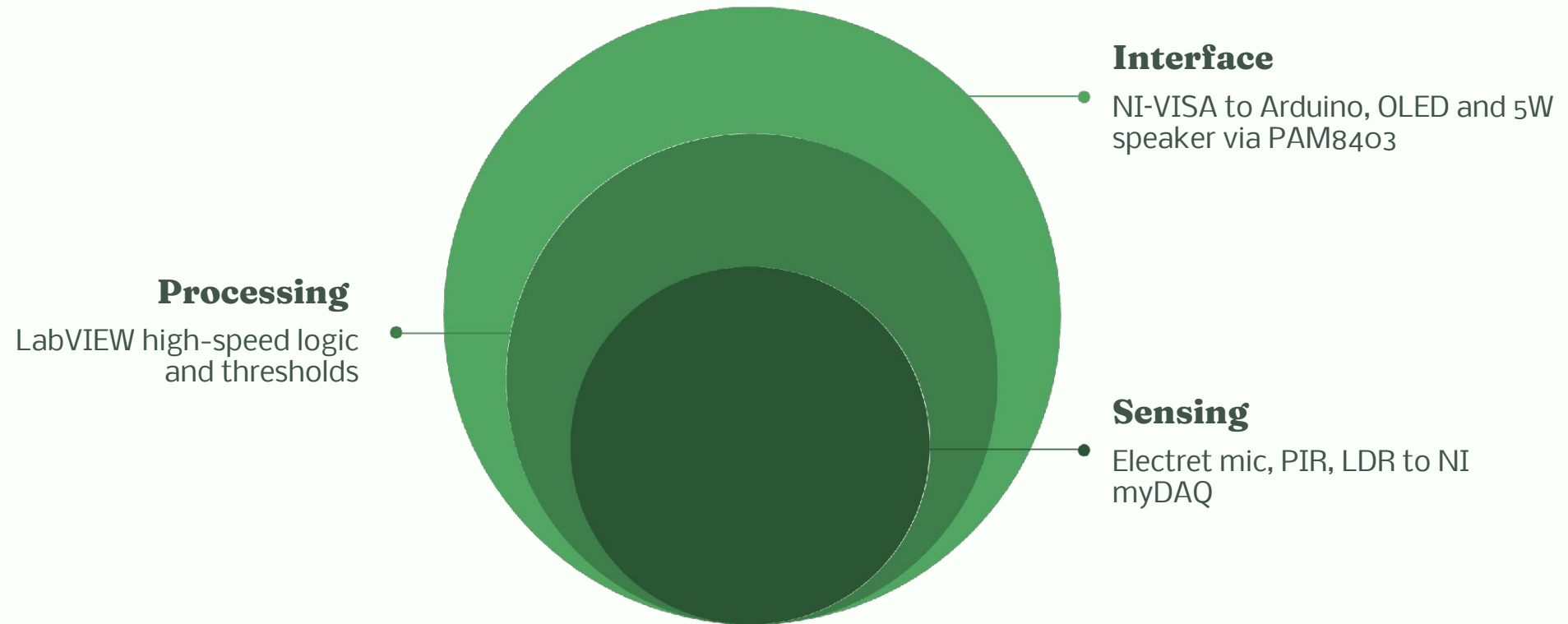
A first-order RC low-pass filter smooths ripple, delivering a **stable DC voltage** proportional to ambient sound intensity – directly readable by the NI myDAQ.



Output: 0 - 5 V DC · Proportional to SPL

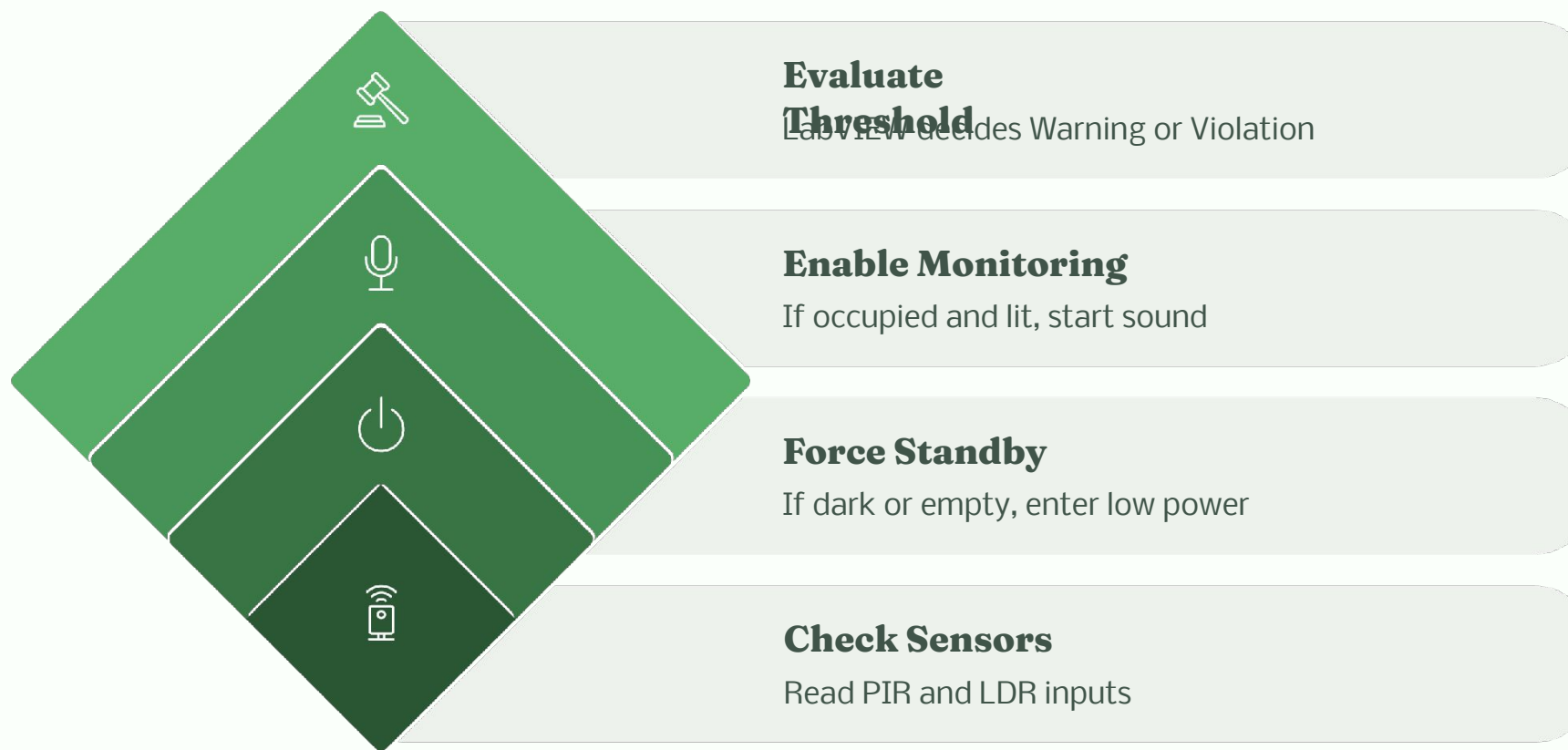


A Decoupled, Three-Layer Data Flow



Separation of concerns protects the DAQ from UI load, ensures deterministic alarm response, and allows each layer to be independently tested and scaled.

Smart Logic & Energy Efficiency — The Gatekeeper



The PIR motion sensor and LDR ambient light sensor act as a hardware gatekeeper. If either detects a dark or unoccupied room, LabVIEW **overrides all sound logic** and forces a low-power Standby state – eliminating false positives and conserving processing resources.

Safe Power Delivery & Audio Amplification

The Hardware Protection Problem

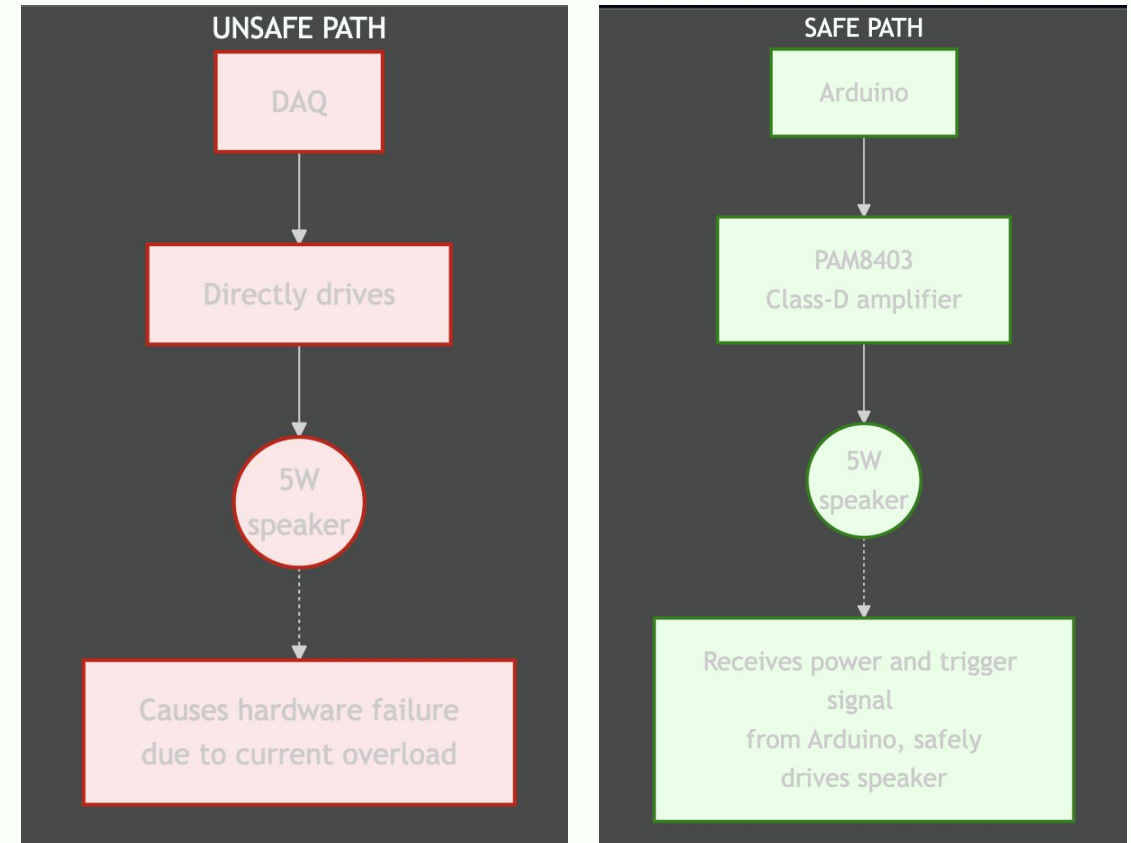
Directly driving a 5 W speaker from the DAQ would exceed current limits, risking irreversible hardware damage.

PAM8403 Class-D Solution

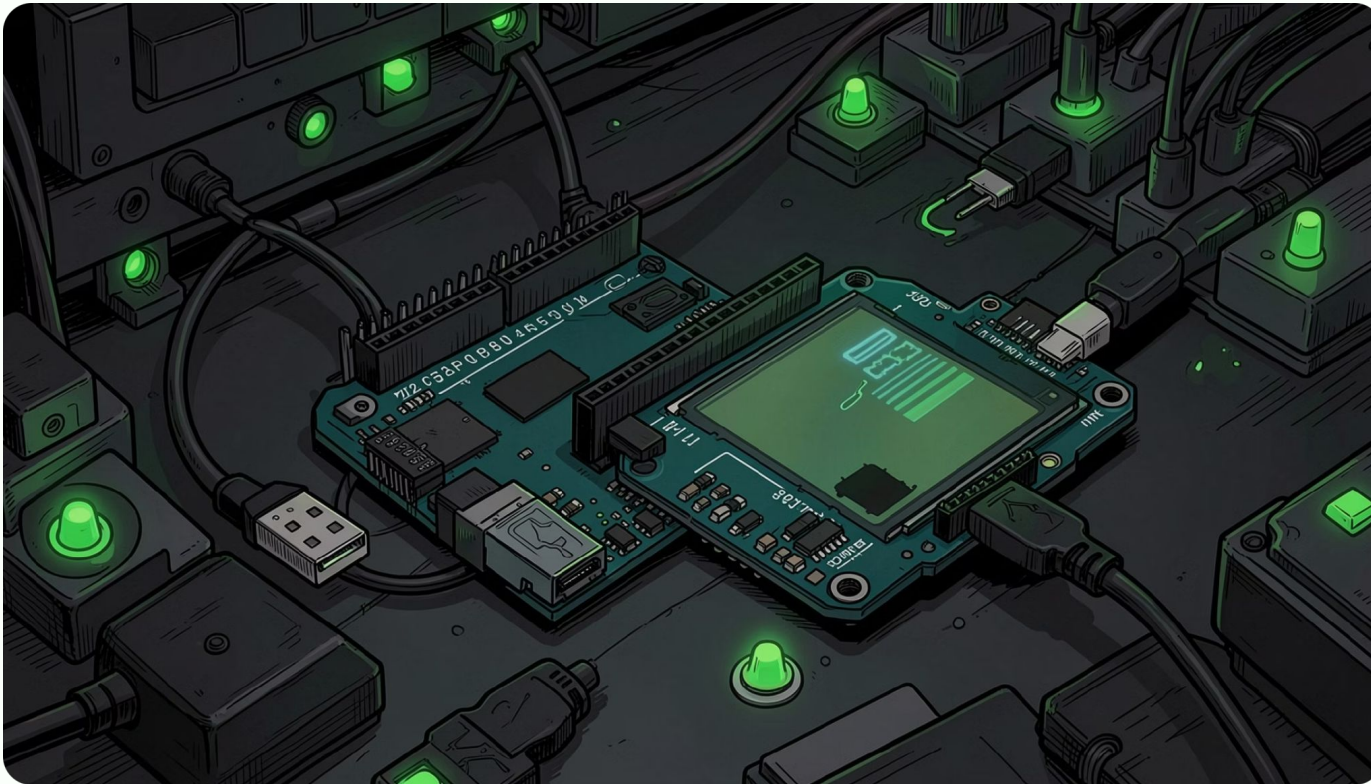
A **PAM8403 Class-D amplifier** draws current from the Arduino – not the DAQ – to safely drive a **1 kHz violation alarm** triggered by the DAQ's GPIO line.



DAQ GPIO current limit: ~8.5mA · Speaker demand: ~1 A



The NI-VISA Serial Bridge

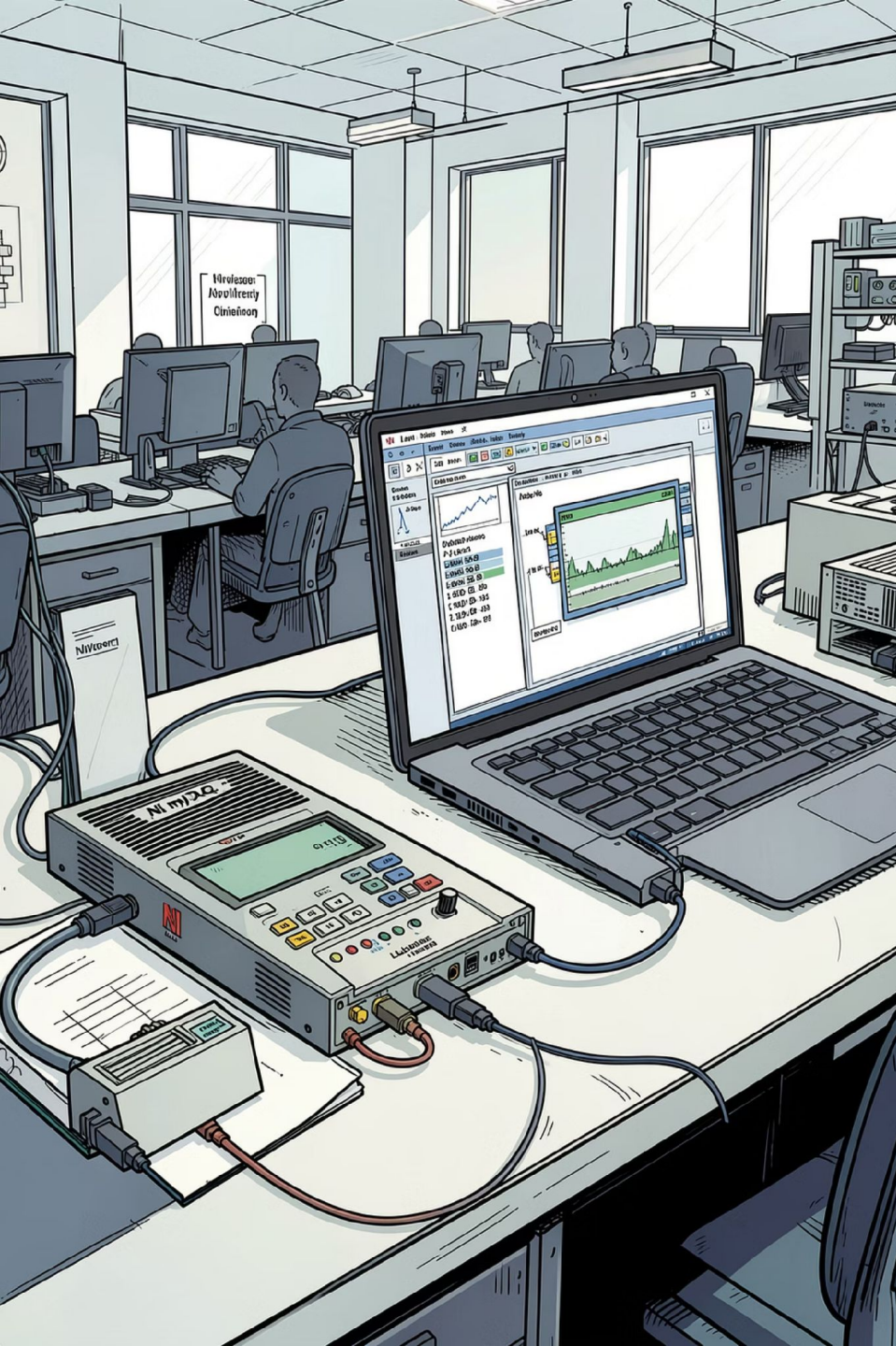


Why Decouple the UI?

Driving the display directly from LabVIEW introduces latency and risks DAQ resource contention. Offloading UI tasks to a dedicated microcontroller preserves acquisition integrity.

Communication Protocol

LabVIEW formats logic states – **Standby, Warning, Violation** – and transmits them via a custom **NI-VISA serial bridge**. The Arduino decodes the buffer and instantly updates a **1.3" I²C OLED display**.



DATA ACQUISITION

LabVIEW Logic & Dynamic Thresholding

Signal Acquisition

The NI myDAQ samples the conditioned analog voltage at high speed, digitizing the signal for real-time LabVIEW processing.

Calibration Routine

On startup, LabVIEW samples the environment for 5 seconds to establish a **baseline ambient noise floor** unique to each deployment location.

Adaptive

Threshold = **Baseline** × **1.1** – a 10% margin above baseline ensures the system adapts to different rooms without manual recalibration.



CONCLUSION

Real-World Viability & Scaling Path

Study Halls & Libraries

Automated, objective enforcement of quiet zones without staff intervention.

Hospital ICUs

Continuous acoustic monitoring to protect patient recovery environments.

Industrial Zones

Scalable deployment for occupational noise compliance and worker safety.



SenseQuiet is an affordable, fully integrated prototype – custom analog front-end, LabVIEW processing, and decoupled UI – ready for commercial scaling.